

Entanglement in Topological Quantum Field Theories and Complexity

Nivedita^a

*^aDepartment of Material Science and Engineering,
Indian Institute of Technology Kanpur,
Uttar Pradesh, India*

E-mail: nivdita@iitk.ac.in

ABSTRACT: The key paper being reviewed is [MMM⁺19] on a new approach to understand Entanglement in Topological Quantum Field Theories.

Contents

1	Introduction	1
2	Knots, Links and Chern- Simons Theory	2
3	Categories and Functors	3
4	Definition of Topological Quantum Field Theory	4
4.1	Axioms of Quantum Mechanics	4
4.2	Definition of a Topological Quantum Field Theory	6
5	Mixed States	7
6	Entanglement Entropy	8
7	Entanglement and Wilson Lines	10
7.1	Arvind's Conjecture	12
8	Complexity	13
8.1	In Quantum Mechanics	13
9	Conclusion	14

1 Introduction

One of the key ideas in quantum mechanics is that two systems that might appear non-interacting may influence each other and behave in ways which are individually random yet are strongly correlated. It is not possible to describe this ‘entanglement’ to be described by any local classical theory [Bel64]. Entanglement is a unique feature of quantum theories which manifests as subtle correlations which exist beyond the limits of naive causality. Let us look at the example of a qubit system. A single qubit is a quantum system has two basis elements $|0\rangle$ and $|1\rangle$. The Hilbert space corresponding to it is $\mathbb{C}^2$. Similarly, for an n -qubit system, the space is $\mathbb{C}^{2^n}$. For three qubits, the basis is $\{|000\rangle, |001\rangle, |010\rangle, |011\rangle, |100\rangle, |101\rangle, |110\rangle, |111\rangle\}$. Let the qubits be A, B and C. To illustrate the idea of entanglement and the relations it has to topology, we look at the specific state, called the GHZ state:

$$|\text{GHZ}\rangle = \frac{|000\rangle + |111\rangle}{\sqrt{2}}$$

On measurement of all three qubits simultaneously, this state results in $|000\rangle$ and $|111\rangle$ with equal probability of 0.5 but if qubit C is made to collapse to either $|0_C\rangle$ or $|1_C\rangle$, the qubits

A and B also collapse to the states $|00\rangle$ or $|11\rangle$ respectively. This idea is similar to what is known as the Borromean rings, which is a three component link with the property that, removing one of the components, un-links the remaining ones. We shall try to understand

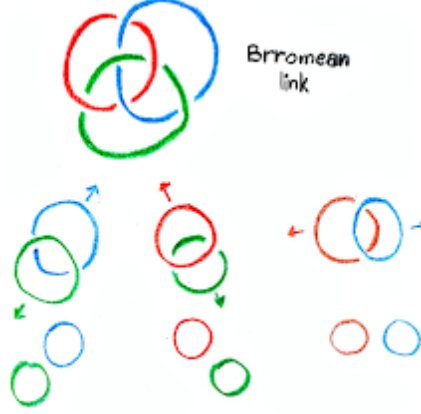


Figure 1. Borromean Link, showing how removing one element unlinks the rest

the connection between topology and entanglement, starting from the modern categorical definition of a Topological Quantum Field Theory.

2 Knots, Links and Chern- Simons Theory

A *knot* is an *embedding* of the circle S^1 into the Euclidean space $\mathbb{R}^3$, or equivalently into the 3-sphere S^3 (one can also consider knots in general 3-manifolds).

A *link* is a *smooth-embedding* of $S^1 \sqcup S^1 \cdots \sqcup S^1 \rightarrow \mathbb{R}^3$ where $S^1 \sqcup S^1 \cdots \sqcup S^1$ denotes a disjoint union of several circles.

Two knots are said to be isotopic (of the same isotopy type) if one can be transformed into another by means of a *smooth isotopy*. A *knot invariant* is map from isotopy equivalence classes of knots to any kind of mathematical structure. These are helpful because it is often much easier to check that the structures one maps to (numbers, groups, polynomials etc.) are different than it is to check that knots are different.

In physics, the starting point for defining a quantum field theory is the choice of a base manifold B of dimension n . The next is the choice of objects over B . In QFT one aims to integrate over the space parametrizing these objects (some kind of a moduli space). These objects are called fields and the operation of integration over the space of fields is also called the path-integral. In integrating over the field space we have to choose a measure on it. The measure is also usually weighted by $\exp(iS)$, where S is a functional on the space of fields in question and is called the action.

Chern-Simons Theory is the most commonly studies topological field theory. The way it connects to mathematical topological or (knot) invariants is as follows:

- The basic strategy to get an invariant out of a QFT goes hand-in-hand with the basic crux of doing the path integral

- We endow the space with extra geometric structures (such as a connection), then compute a number from this manifold-with-connection (using the action functional)
- Finally we sum over all geometric structures (in this case, equivalence classes of connections) to take away the dependence on the added structure

The final result is a topological invariant because all the non-topological (geometric) data has been integrated over.

We use the Chern- Simons action because it does not have any dependence on a metric and the connection is the only geometric quantity is the entire structure which we later perform the path integral over.

We take the Chern-Simons action as:

$$S_{\text{CS}}[A] = \frac{k}{4\pi} \int_B dA \wedge A + \frac{2}{3} A \wedge A \wedge A$$

The path integral is given by

$$Z_B(k) = \int \mathcal{D}A e^{iS_{\text{CS}}[A]}$$

Here, A is a Lie algebra-valued gauge field, Also, if we have a probe particle, an observable quantity which is dependent on the topology of the system, *holonomy* which encodes the information of transporting the probe particle along the given path under the influence of the field.

Chern-Simons theory or in fact any topological theory is characterised by triviality of the TQFT Hamiltonian, since T^{00} component of the stress energy tensor represents Hamiltonian density and,

$$T^{ij} \sim \frac{\delta S}{\delta g_{ij}} = 0$$

Thus, TQFTs do not have non-trivial local dynamics. Non-local correlations arise from topology of the base manifold or from singularities in the field itself (defects).

3 Categories and Functors

A *category* $\mathcal{C}$ consists of the following:

- a collection of *objects*
- a collection of *morphisms*

such that,

- Each morphism has specified domain and codomain objects, $f: X \rightarrow Y$ signifies f is a morphism with domain X and codomain Y where X and Y are objects of the category
- each object has a designated *identity* morphism $1_X: X \rightarrow X$.

- For any pair of morphisms f, g with the codomain of f equal to the domain of g , there exists a specified *composite morphism* $g \circ f$

This data is subject to two axioms:

- For any $f: X \rightarrow Y$, the composites $1_Y \circ f$ and $f \circ 1_X$ are both equal to f
- For any composable triple of morphisms f, g, h the composites $h \circ (g \circ f)$ and $(h \circ g) \circ f$ are equal, associativity

Examples: Some commonly studied categories are **Set** with objects as sets and morphisms as functions, **Grp** of groups and morphisms as group homomorphisms, **Man** as smooth manifolds as objects and smooth (infinitely differentiable) maps as morphisms. In the present context we will study a category called **Cob_n** or the Cobordism Category.

Next, we talk about a *functors*. It is said by John Baez in ‘Quantum Quandaries’ that *... every sufficiently good analogy is yearning to become a functor*. So to move from spaces and manifolds to abstract Hilbert spaces of Quantum Mechanics we will need a functor. Let $\mathcal{C}$ and $\mathcal{D}$ be categories. A *functor* $F: \mathcal{C} \rightarrow \mathcal{D}$ consists of the data:

- A function $F: \text{ob } \mathcal{C} \rightarrow \text{ob } \mathcal{D}$ that is the objects of one category are mapped to objects of the other category
- A function $F: \mathcal{C}[A, B] \rightarrow \mathcal{D}[FA, FB]$ for every pair of objects A, B in $\mathcal{C}$, and morphisms between objects are also mapped to morphisms between images of respective objects under the functor

For F to be a functor, it must satisfy the following properties:

- For any $C \in \mathcal{C}$ we have $F(1_C) = 1_{FC}$
- For any two morphisms f, g we have $F(f \circ g) = Ff \circ Fg$ whenever $f \circ g$ is defined.

We call a category monoidal if there is a notion of tensor product, that is, a functor

$$\otimes: \mathcal{C} \times \mathcal{C} \rightarrow \mathcal{C}$$

such that the $\otimes$ is associative and has a unit object $I \in \text{ob } \mathcal{C}$. A functor that preserves these tensor products, that is, a functor such that $F(f \otimes g) \simeq F(f) \otimes F(g)$ for both objects and morphisms and also preserves the init object $F(I_C) \simeq I_D$

4 Definition of Topological Quantum Field Theory

4.1 Axioms of Quantum Mechanics

The axioms of quantum mechanics are

1. A complex Hilbert space $\mathcal{H}$ is assigned to the system under consideration

2. The space of states (pure states) of a physical system is given by rays (rays, because we only need state vectors up to the equivalence of being multiplied by a complex number) in a Hilbert space $\mathcal{H}$
3. Observables are represented by Hermitian operators on $\mathcal{H}$
4. If we measure the observable $\mathcal{O}$ in a state $|\psi\rangle$, then the outcome is always an eigenvalue of $\mathcal{O}$. The probability of getting eigenvalue ω is given by

$$\frac{\langle\psi|\mathbb{P}_\omega|\psi\rangle}{\langle\psi|\psi\rangle}$$

where $\mathbb{P}_\omega$ is the projection operator onto the eigenspace of ω . On measurement giving the result ω , the state ψ collapses into $\mathbb{P}_\omega|\psi\rangle$

5. Time evolution is given by unitary operators, that is the time evolution from t_1 to t_2 is given by a unitary map

$$U(t_1 \rightarrow t_2) : \mathcal{H} \rightarrow \mathcal{H}$$

These maps satisfy the following locality condition

$$U(t_2 \rightarrow t_3) \circ U(t_1 \rightarrow t_2) = U(t_1 \rightarrow t_3)$$

and $U(t, t) = \text{id}_{\mathcal{H}}$. We also assume U depends smoothly on t_1, t_2

6. Given independent subsystems of a system with Hilbert spaces $\mathcal{H}_i$ for $i \in I$, the Hilbert space of the total system is

$$\bigotimes_{i \in I} \mathcal{H}_i$$

We will now rephrase axioms 1, 5, 6 to make the categorical structure of quantum mechanics apparent. We assume that $U(t_1, t_2)$ depends only on $t_2 - t_1$ as a fundamental theory of physics is expected to be time-translation invariant. We define the category of $\mathbf{Cob}_1^{\text{Riem}}$ of 1 dimensional Riemannian cobordisms (having a metric or a notion of length). Its objects are 0-manifolds (finite or countable collections of points). Given two 0-manifolds with cardinality n and m (as 0-manifolds are completely determined by cardinality and an assignment of orientation to each point), an arrow $t: n \rightarrow m$ is a oriented compact Riemannian 1-manifold upto isometry such that:

$$\partial t = \bar{n} \sqcup m$$

We call the 1- manifold t , because it signifies time evolution of the point. Composition is given by gluing of endpoints. Let $\mathbf{Hilb}_{\mathbb{C}}^{\text{unit}}$ be the category with objects as complex Hilbert spaces and arrows being unitary maps. Both categories have monoidal structure, $\sqcup$ (disjoint union) and $\otimes$ (tensor product) on $\mathbf{Cob}_1^{\text{Riem}}$ and $\mathbf{Hilb}_{\mathbb{C}}^{\text{unit}}$ respectively. Then axioms 1, 4, 5 basically say that a quantum mechanical theory is a strong monoidal functor

$$Z: \mathbf{Cob}_1^{\text{Riem}} \rightarrow \mathbf{Hilb}_{\mathbb{C}}^{\text{unit}}$$

This is how we recover the axioms:

- All 0-manifolds can be obtained by the disjoint union of a collection of points. So by the fact that the functor preserves tensor product, we see that $Z(\cdot^+)$ and $Z(\cdot^-)$ determine Z on all objects. By seeing the interval $[0, 1]$ as a cobordism $\cdot^+ \sqcup \cdot^- \rightarrow \emptyset$, we get a map $Z(\cdot^+) \otimes Z(\cdot^-) \rightarrow \mathbb{C}$. This shows that $Z(\cdot^-) = Z(\cdot^+)^*$. So $Z(\cdot)$ determines Z on all objects. Define

$$\mathcal{H} = Z(\cdot)$$

Hence we get axiom 1 that the system is assigned a Hilbert space.

- Up to isometry (length preserving maps), the set of arrows from $\cdot^+ \rightarrow \cdot^+$ is given by the intervals $[0, t] = I_t$. Define $U(t_0, t_0 + t) = Z(I_t)$. Then

$$U(t_0 + t, t_0 + t') \circ U(t_0, t_0 + t) = Z(I_{t'}) \circ Z(I_t) = Z(I_{t+t'}) = U(t_0, t_0 + t + t')$$

where we have used time-translation invariance. Also, $Z(I_0) = \text{id}_{Z(\cdot^+)}$. The previous equations follow from the properties of functors. This verifies axiom 5 that time evolution is a unitary operator.

- Axiom 6 follows obviously from the fact that Z is a strong monoidal functor, i.e. multi-component systems or many particle systems can be assigned tensor products of Hilbert Spaces.

We still require axioms 2, 3, 4 to interpret the quantum mechanical theory we obtain from Z , which define state

4.2 Definition of a Topological Quantum Field Theory

We can try to extend the above definition to Higher Dimensions and define a Quantum Field Theory to be a strong monoidal functor from a higher dimensional category of Riemannian cobordisms $\mathbf{Cob}_n^{\text{Riem}}$ to $\mathbf{Hilb}_{\mathbb{C}}^{\text{unit}}$. However, the category of Riemannian cobordisms is hard to define in higher dimensions (identity morphisms cannot be defined in the right way) so we simplify. We define the category $\mathbf{Cob}_n$ of smooth cobordisms to have objects as compact oriented smooth $n - 1$ -manifolds and we define an arrow $M: \Sigma \rightarrow \Sigma'$ to be a compact oriented smooth n -manifold (up to diffeomorphism) such that

$$\partial M = \bar{\Sigma} \sqcup \Sigma'$$

Once again this is a monoidal category under $\sqcup$ with unit $\emptyset$. We define a n -dimensional *topological quantum field theory*, following Atiyah [Ati88] to be a strong monoidal functor

$$Z: \mathbf{Cob}_n \rightarrow \mathbf{Vect}_{\mathbb{C}}$$

where $\mathbf{Vect}_{\mathbb{C}}$ is the category of complex vector spaces and linear maps with the obvious monoidal structure. This is a topological theory as the cobordisms are now no longer Riemannian manifolds but smooth (not equipped with a metric) and are quotiented by diffeomorphism, not isometry. We replace $\mathbf{Hilb}_{\mathbb{C}}^{\text{unit}}$ by $\mathbf{Vect}_{\mathbb{C}}$ to allow for more complex non-unitary theories. So facts that this definition entails are

- $Z(M \sqcup N) \simeq Z(M) \otimes Z(N)$ and $Z(\emptyset) = \mathbb{C}$. These two follow trivially from the fact definition of strong monoidal functor
- $Z(\overline{M}) = Z(M)^*$. This follows in a similar way as in the quantum mechanical case.
- $Z(\Sigma \times I) = \text{id}_{Z(\Sigma)}$. This follows from the fact that $\Sigma \times I$ is the identity morphisms $\text{id}_\Sigma: \Sigma \rightarrow \Sigma$ in $\mathbf{Cob}_n$.
- Given an n -manifold M such that $\partial M = \Sigma_1 \sqcup \Sigma_2$ we can think of it as a cobordism in two ways. First $M_1: \overline{\Sigma}_1 \rightarrow \Sigma_2$ and $M_2: \overline{\Sigma}_2 \rightarrow \Sigma_1$. Then

$$\begin{aligned} Z(M_1) &\in \text{Hom}(Z(\Sigma_1)^*, Z(\Sigma_2)) \\ Z(M_2) &\in \text{Hom}(Z(\Sigma_2)^*, Z(\Sigma_1)) \end{aligned}$$

$Z(M_1), Z(M_2)$ are related by the natural isomorphisms $\text{Hom}(V, W) \simeq \text{Hom}(W^*, V^*)$ and $V \simeq V^{**}$.

- In the previous point, we can choose $\Sigma_1 = \partial M, \Sigma_2 = \emptyset$. Then $Z(M_2): \mathbb{C} \rightarrow Z(\partial M)$. Using the natural isomorphism $\text{Hom}(\mathbb{C}, V) \simeq V$ we can think of $Z(M_2) \in Z(\partial M)$. We will write $Z(M_2) = |M\rangle$. Similarly, $Z((\overline{M})_2) \in Z(\overline{\partial M}) \simeq Z(\partial M)^*$. We write $Z((\overline{M})_2) = \langle M|$.
- Given two manifold M, N such that $\partial M \simeq \partial N \simeq \Sigma$ we can form the states $|M\rangle \in Z(\Sigma)$ and $\langle N| \in Z(\Sigma)^*$. So using the natural evaluation map $V^* \otimes V \rightarrow \mathbb{C}$, we obtain a complex number $\langle N|M\rangle$.
- For a closed n -manifold M ($\partial M = \emptyset$), we can view it as a cobordism $M_3: \emptyset \rightarrow \emptyset$. So $Z(M): \mathbb{C} \rightarrow \mathbb{C}$. Using the natural isomorphism $\text{Hom}(\mathbb{C}, V) \simeq V$, we can think of $Z(M) \in \mathbb{C}$. This is called the partition function on a closed manifold.
-

5 Mixed States

We call a unit norm vector in the Hilbert space a *pure state*. Given any pure state we can define an operator

$$\rho_\psi = \frac{|\psi\rangle \otimes \langle\psi|}{\langle\psi|\psi\rangle}$$

In fact, ρ_ψ is just the projection operator onto the ray spanned by $|\psi\rangle$. Notice that ρ_ψ is positive semidefinite and has trace 1

$$\begin{aligned} \langle\phi|\rho_\psi|\phi\rangle &= \frac{|\langle\phi|\psi\rangle|^2}{\langle\psi|\psi\rangle} \\ \text{tr}(\rho_\psi) &= \text{tr}\left(\frac{|\psi\rangle \otimes \langle\psi|}{\langle\psi|\psi\rangle}\right) = \text{tr}\left(\frac{\langle\psi|\psi\rangle}{\langle\psi|\psi\rangle}\right) = \text{tr}(\text{id}_\mathbb{C}) = 1 \end{aligned}$$

We define any positive semi-definite Hermitian operator of trace 1 to be a *mixed state*.

We have $\rho_\psi |\psi\rangle = |\psi\rangle$. So one of ρ_ψ 's eigenvalues is 1 and all it's eigenvalues are ≥ 0 (positive semi definite). Since all eigenvalues must add to 1 (trace is 1) it must have all other eigenvalues to be 0. But such a matrix will necessarily be a projection onto a 1 dimensional subspace of $\mathcal{H}$, and thus we can recover a unique ray from a a mixed state with one its eigenvalues 1 and others 0. So we see that pure states are in bijection with those mixed states with only one non-vanishing eigenvalue (which is necessarily 1). Mixed states corresponding to pure states also satisfy $\rho^2 = \rho$.

For any mixed state ρ , we can express it as

$$\rho = \sum_i p_i |b_i\rangle \langle b_i|$$

where b_i is an orthonormal eigen basis for ρ and $0 \leq p_i \leq 1$ and $\sum_i p_i = 1$. Let $\mathcal{O}$ be any operator, then

$$\text{tr}(\rho\mathcal{O}) = \sum_i p_i \text{tr}(|b_i\rangle \langle b_i| \mathcal{O}) = \sum_i p_i \langle b_i| \mathcal{O} |b_i\rangle$$

From here we can see the physical meaning of a mixed state ρ . The $\langle b_i| \mathcal{O} |b_i\rangle$ terms are quantum expectation values. If we interpret p_i to be a probability distribution over the b_i , then the sum can be thought of as an extra (non-quantum) averaging. Such a situation can arise if we have a lack of knowledge about the true quantum state of the system (for example in statistical mechanics, or as we shall see, if we forget a part of an entangled system).

6 Entanglement Entropy

Entanglement Entropy is the entropy for an observer who can only access a part of the total system, i.e. some subsystem. It is a measures of **amount of quantum information**. It can also be viewed as a non-local version of the correlation functions as it captures the global data like the topological information about the system. Thus, it can be used as a **quantum order parameter**. Another way to think of Entanglement Entropy is that it is related to the degrees of freedom of the system, non vanishing even at zero temperature thus it becomes a useful parameter to study in quantum many-body systems. This concept precisely is the bridge between gravity and condensed matter physics.

To precisely define Entanglement Entropy, we go about the following way, Divide the entire system into two subsystems A and B . Then the total Hilbert Space gets factorised to $\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_B$. Then as stated above we can define the total density matrix ρ , similarly next we can define the reduced density matrix ρ_A for A as $\rho_A = \text{Tr}_B \rho$. That is we trace over the subsystem B . Finally the Entanglement Entropy or Von-Neumann Entropy is defined as

$$S_A = -\text{Tr}_A \rho_A \log \rho_A$$

The standard thermal entropy is just a special case of this where we take A to be total system. A basic trick to calculate entanglement Entropy in QFT is the **replica method**. Let λ_i be the eigenvalues of ρ_A Then, $\log \text{Tr}(\rho_A)^n = \log(\sum_i \lambda_i^n)$

$$-\frac{\partial}{\partial n} \log \text{Tr}(\rho_A)^n \big|_{n=1} = -\frac{\sum_i \lambda_i^n \log \lambda_i}{\sum_i \lambda_i^n} \big|_{n=1} = -\frac{\sum_i \lambda_i \log \lambda_i}{\sum_i \lambda_i} = -\sum_i \lambda_i \log \lambda_i = -\text{Tr} \rho_A \log \rho_A$$

One can also see that $S_A = -\frac{\partial}{\partial n} \log \text{Tr}(\rho_A)^n \big|_{n=1} = -\frac{\partial}{\partial n} \text{Tr}(\rho_A)^n \big|_{n=1}$ because $\text{Tr} \rho_A = 1$ by definition. Replica trick is a very natural tool in TQFT: first, the density matrix can be interpreted as a cobordism, and, the matrix multiplication is straightforward, since it is a natural composition of cobordisms or gluing along the common boundary.

Example:-

$$\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_B$$

The partition function [Z functor] would have a disjoint union of spaces $\Sigma_A \sqcup \Sigma_B$ such that (n-1 dimensional) in domain category

$$Z[\Sigma_A \sqcup \Sigma_B] = Z[\Sigma_A] \otimes Z[\Sigma_B] = \mathcal{H}_A \otimes \mathcal{H}_B$$

Two Candidates:-

$$|\psi_1\rangle = Z \left[\begin{array}{c} \Sigma_A \\ \Sigma_B \end{array} \right]$$

$$|\psi_2\rangle = Z \left[\begin{array}{c} \Sigma_A \quad \Sigma_B \end{array} \right]$$

$$\hat{P}_1 = Z \left[\begin{array}{cc} \Sigma_A & \Sigma_B \\ \Sigma_A & \Sigma_B \end{array} \right]$$

$$\hat{P}_2 = Z \left[\begin{array}{c} \Sigma_A \quad \Sigma_B \\ \Sigma_A \quad \Sigma_B \end{array} \right]$$

Normalization - Dividing by Trace
Trace - gluing/contraction of a pair of indices

$$\text{Tr} \hat{P}_1 = Z \left[\begin{array}{c} \text{Sphere} \end{array} \right]^2$$

$$\text{Tr} \hat{P}_2 = Z \left[\begin{array}{c} \text{Torus} \end{array} \right]$$

$$e_1 = (\text{Tr} \hat{P}_1)^{-1} \hat{P}_1$$

$$P_2 = (\text{Tr} P_2)^{-1} \hat{P}_2$$

Reduced density matrix: Trace over subsystem B

$$P_1^A = \frac{Z \left[\begin{array}{c} \Sigma_A \\ \Sigma_B \end{array} \right] Z \left[\begin{array}{c} \Sigma_B \end{array} \right]}{Z \left[\begin{array}{c} \text{Sphere} \end{array} \right]^2}$$

$$P_2^A = \frac{Z \left[\begin{array}{c} \Sigma_A \\ \Sigma_B \end{array} \right]}{Z \left[\begin{array}{c} \text{Torus} \end{array} \right]}$$

Replica Trick of computing Entropy $S = -\lim_{n \rightarrow 1} \frac{d \ln [\text{Tr}_A (P_1^A)^n]}{dn}$

$$\begin{aligned}
(P_1^A)^n &= \frac{1}{Z[\text{circle}]} Z \left[\overbrace{\text{cup} \text{cup} \dots \text{cup}}^n \right] \\
&= \frac{Z[\text{cup}] Z[\text{circle}]^{n-1}}{Z[\text{circle}]^n} \\
&= \frac{Z[\text{cup}]}{Z[\text{circle}]} \\
\text{Tr}(P_1^A)^n &= 1
\end{aligned}$$

$$S_1 = 0$$

$$\begin{aligned}
(P_2^A)^n &= \frac{1}{Z[\text{circle}]} Z \left[\overbrace{\text{cup} \text{cup} \dots \text{cup}}^n \right]_{\text{glue}} \\
&= \frac{Z[\text{cup}]}{Z[\text{circle}]^n}
\end{aligned}$$

$$\text{Tr}(P_2^A)^n = Z[\text{cup}] / Z[\text{circle}]^n = Z[\text{cup}]^{1-n}$$

$$S_2 = -\frac{d}{dn} (1-n) \log Z[\text{cup}]$$

$$S_2 = \log Z[\text{cup}] \neq 0 \text{ in general (depends on theory and exact form of } Z)$$

7 Entanglement and Wilson Lines

A Riemann surface is a one dimensional complex manifold or a 2-D surface. If there are punctures in a Riemann surface, it is homeomorphic to a surface with boundaries. Now,

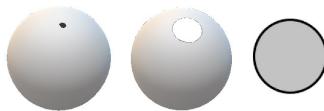


Figure 2. A sphere with a puncture, a sphere with a disk cut out and a disk are equivalent

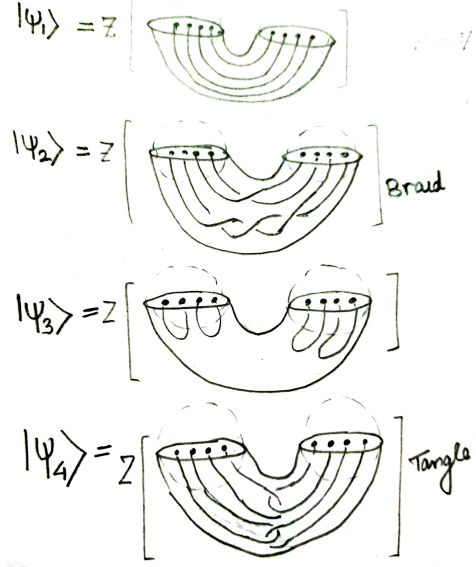


Figure 3. The various states possible with the cobordism going from S^2 to S^2 with 4 punctures

we consider cobordisms whose boundaries have punctures and the punctures are connected by strings running through the cobordism referred to as ‘Wilson lines’. The gluing of boundaries or composition is only possible if the boundaries are cobordant or similar and the punctures (thought of as boundaries of the boundary) are also cobordant.

As an example we shall consider S^2 as the boundary manifold. From CFT (considering punctures as defects), it is clear that for having a non trivial Hilbert space, at least three (or more punctures) are needed otherwise there is no entanglement possible. We consider the following states, We conclude the following:

- $|\psi_1\rangle$ The entropy is the logarithm of the dimension of the Hilbert space associated with S^2 with n punctures and this is greater than 1.
- $|\psi_2\rangle$ Braiding of Wilson lines does not affect the entropy, since it amounts to a local operation on either of the spheres, of which the von Neumann entropy is independent, since this operation corresponds to a local unitary operation and because entropy includes taking a trace, it doesn’t get affected. This can also be seen explicitly by drawing the lines.
- $|\psi_3\rangle$ $\text{Tr} \rho^n$ will be expressed in terms of the link invariant of un-linked circles in $S^2 \times S^1$ that do not wind around S^1 . (Note that we get S^1 on tracing over the Identity or Interval as the original manifold was $S^2 \times I$. On normalization we get 1, log of which gives zero.
- $|\psi_4\rangle$ Lastly, This needs evaluation of invariants of chain with n segments of links in $S^2 \times S^1$, this computation is still a topic of ongoing research.

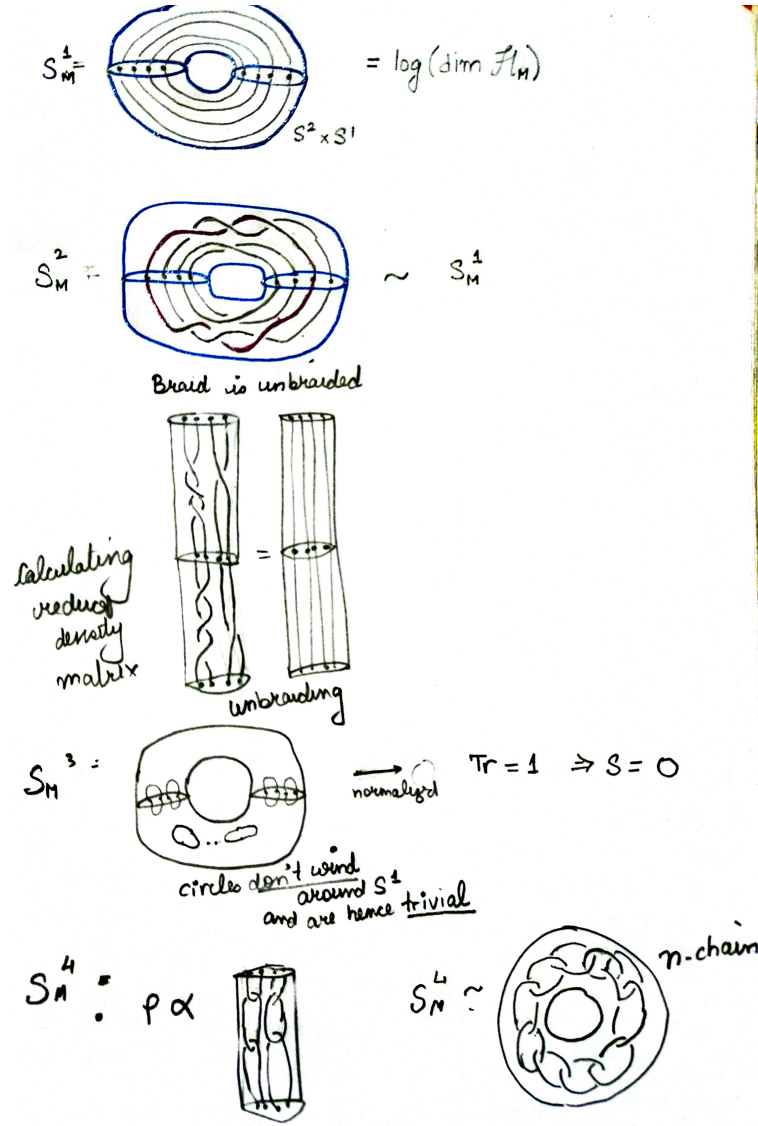


Figure 4. The various states possible with the corbordism going from S^2 to S^2 with 4 punctures

7.1 Arvind's Conjecture

As stated right in the beginning of the report, a proposal exists in terms of interpreting entangled states in terms of links. To do so, we consider a state in a TQFT obtained by cutting out solid tubular neighbourhoods (cylinders) around a link inside a 3 dimensional sphere S^3 , what we are left with is a space homeomorphic to a solid torus. The complement of an n -component link is a three dimensional space $\mathcal{M}_3$ whose boundary is disjoint union of n -tori in general. Some conclusions that are made by studying the entanglement entropy of different links in S^3 are as follows under Chern Simons theory:

- The link is trivial if and only if Von Neumann entropy associated to it vanishes, as the partition function factorizes.

- Hopf link (2- component) and is analogous to a Bell pair.

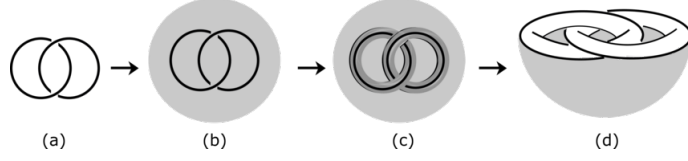


Figure 5. This image is taken from reference [DSD⁺17] and shows a Hopf link is first thickened into a tubular neighbourhood and then removed. The resulting manifold $S^3 \setminus H$ has a boundary consisting of two linked tori $T^2 \sqcup T^2$. Our formulation for TQFT gives a quantum state on the boundary whose entanglement structure can be studied.

The resulting Hilbert space would be $\mathcal{H}_{T^2} \otimes \mathcal{H}_{T^2}$ and any state written as $\sum_{a,b} C_{ab} |a\rangle \otimes |b\rangle$. The exact evaluation of the entropy in Chern Simons theory depends on the level k , the rank r and the number of integrable representations $\mathcal{I}$ of the lie algebra. Various limits of this have been calculated in literature and case of the above example turns out to be equal to the upper bound, hence maximally entangled. This takes us forward in Arvind’s conjecture in the sense that entanglement implies linking but no one to one correspondence is yet established.

8 Complexity

Computations come with a notion of complexity of the given task usually in terms of time (time complexity) and resources (space complexity). In QFT we have a notion of complexity imported, called circuit complexity from theoretical computer science [AC19], here we shall see that approach discussed in terms of quantum mechanics.

8.1 In Quantum Mechanics

In quantum mechanics it is an important consideration as to how to go from one state of the system to another. In other we talk about to states a ‘target’ state $|\psi_T\rangle$ to be prepared and a given ‘reference state’ $|\psi_R\rangle$. We can write the following equation for some unitary operator U ,

$$|\psi_T\rangle = U |\psi_R\rangle$$

It can be shown that for each finite dimensional Hilbert Space, there exist a finite set of unitary operators U_a such that any other unitary can be approximated to arbitrary accuracy by products of the U_a . This set of U_a are called basic gates.

A quantum circuit connecting two states is a a ordered list of basic gates which product to the desired unitary (up to a certain accuracy).

The number of U_a in a given circuit C is called its **depth** defined as $D(C)$, and complexity of a unitary is the minimum depth required to simulate it.

$$\mathcal{C}(\psi_R, \psi_T; \{U_a\}) = \min_{C=\{U_a\} | \prod U_a \approx U} D(C)$$

The circuit can be thought of as time evolving the state with a time dependent Hamiltonian. We can take U_a to be the time evolution with an operator $\mathcal{O}_a$ such that $U_a = \exp(\epsilon \mathcal{O})$

$$U(t) = \mathcal{P} \exp \left(i \int_0^t H(t) dt \right) \text{ where } H(t) = \sum_a Y^a(t) \mathcal{O}_a$$

where we have introduced the function $Y^a(t)$ which decides whether or not the operator U^a is applied at the level t of the circuit and we also define $U(0) = 1$. Also, the path ordering is important since the operators at different levels s of Hamiltonian at different times may not commute.

If we consider the case where the Hamiltonian itself is Lie algebra valued, that is, the $\mathcal{O}_a = \mathcal{O}_{ai} T^i$, where T^i are the basis of the lie-algebra. The

The path ordered exponential then can be cast in terms of paths on the group space (because exponentiating the lie algebra elements takes us to the corresponding Lie group), that is the space of available unitary operators. Complexity $\mathcal{C}$ in such a case can be defined as the minimum length, or the geodesic distance, of the path in the group space (Lie Group Manifold) connecting the identity element and the operator $U(t)$.

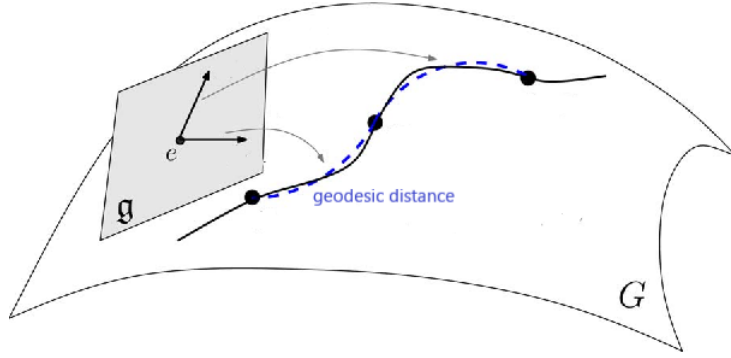


Figure 6. Exponential map from lie algebra to its lie group, note that Lie Algebra can be thought of as the tangent space at the e identity element of the Lie Group

9 Conclusion

In this review we looked at the new mathematical formulations of TQFT, defined the concept of mixed states, entanglement and entanglement entropy. We showed by demonstration of a calculation using the replica technique. Next we performed the same calculation but this time having punctures in the boundaries which are connected by Wilson lines. We observed that links can be assigned physical interpretation in terms of entropy. Finally, we talked about the notion of complexity and its geometric interpretation.

References

- [AC19] Daniel W. F. Alves and Giancarlo Camilo. Momentum-space entanglement after smooth quenches. *The European Physical Journal C*, 79(1):48, Jan 2019.
- [Ati88] Michael F. Atiyah. Topological quantum field theories. *Publications Mathématiques de l'Institut des Hautes Études Scientifiques*, 68(1):175–186, January 1988.
- [Bel64] J. S. Bell. On the einstein podolsky rosen paradox. *Physics Physique Fizika*, 1:195–200, Nov 1964.
- [DSD⁺17] Siddharth Dwivedi, Vivek Singh, Saswati Dhara, P. Ramadevi, Yang Zhou, and Lata Joshi. Entanglement on linked boundaries in chern-simons theory with generic gauge groups. *Journal of High Energy Physics*, 2018, 11 2017.
- [MMM⁺19] D. Melnikov, A. Mironov, S. Mironov, A. Morozov, and An. Morozov. From topological to quantum entanglement. *Journal of High Energy Physics*, 2019(5):116, May 2019.